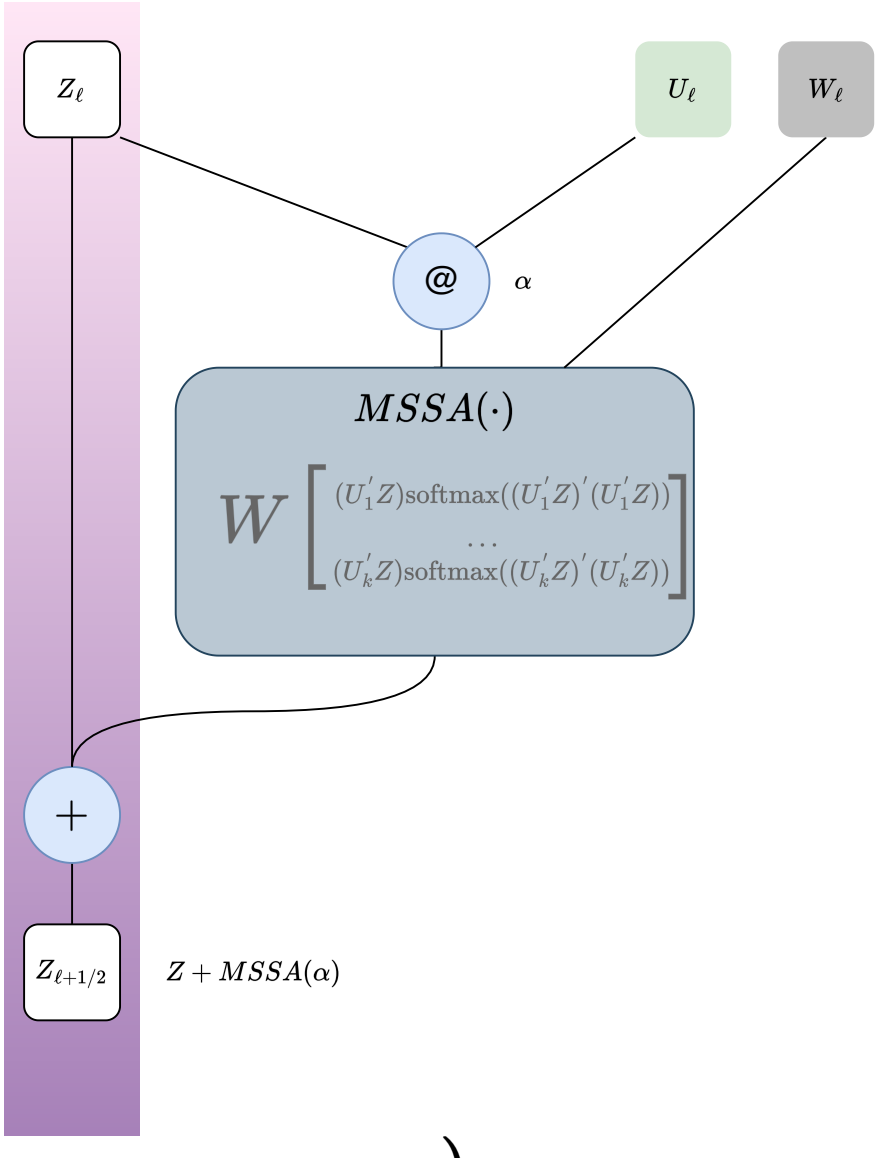
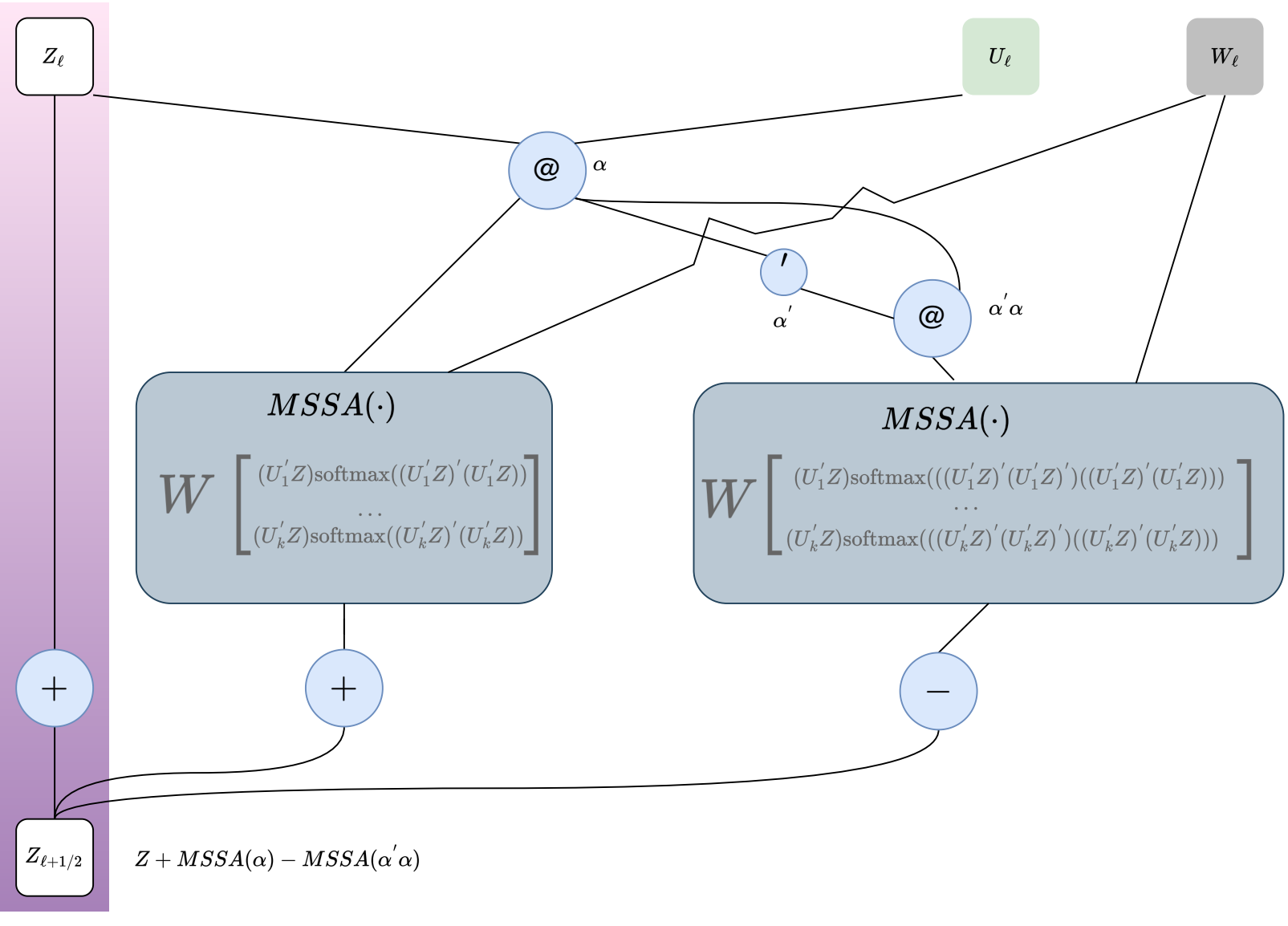

Z_ℓ
U_ℓ
W_ℓ
@
α
$MSSA(\cdot)$
$W \begin{bmatrix} (U_1'Z)\text{softmax}((U_1'Z)'(U_1'Z)) \\ \cdots \\ (U_k'Z)\text{softmax}((U_k'Z)'(U_k'Z)) \end{bmatrix}$
+
$Z_{\ell+1/2}$
$Z + MSSA(\alpha)$

a)


Z_ℓ
U_ℓ
W_ℓ
@
α
'
α'
@
$\alpha'\alpha$
$MSSA(\cdot)$
$W \begin{bmatrix} (U_1'Z)\text{softmax}((U_1'Z)'(U_1'Z)) \\ \cdots \\ (U_k'Z)\text{softmax}((U_k'Z)'(U_k'Z)) \end{bmatrix}$
$MSSA(\cdot)$
$W \begin{bmatrix} (U_1'Z)\text{softmax}(((U_1'Z)'(U_1'Z)')((U_1'Z)'(U_1'Z))) \\ \cdots \\ (U_k'Z)\text{softmax}(((U_k'Z)'(U_k'Z)')((U_k'Z)'(U_k'Z))) \end{bmatrix}$
+
+
−
$Z_{\ell+1/2}$
$Z + MSSA(\alpha) - MSSA(\alpha'\alpha)$

b)